

Simulation of Lid Driven Cavity flow using SIMPLE algorithm

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1 Introduction

In this report we will try to implement SIMPLE algorithm for a lid driven cavity flow and try to validate results from benchmark results.

2 Mesh/Grid generation

We are taking a simple equidistant mesh/grid to solve the lid driven cavity flow in 2D domain. The fundamental idea behind the SIMPLE algorithm is to make velocity staggered over pressure.

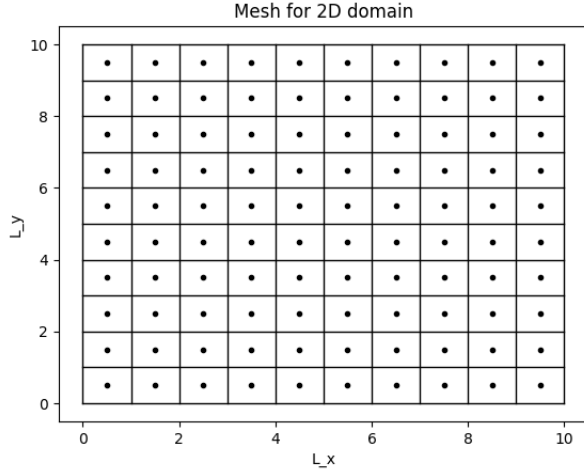


Figure 1: Collocated mesh

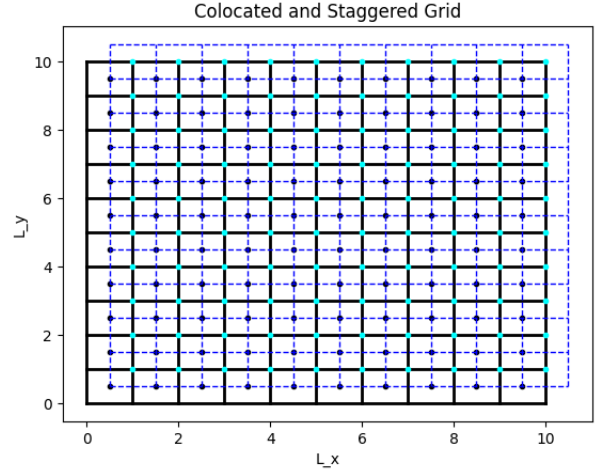


Figure 2: Staggered mesh

The concept of staggered mesh is to avoid checker boarding issue. We will first look at checker boarding issue and how it can be resolved using staggered mesh.

- Checker Boarding issue : Consider a velocity field over a collocated grid, in continuity equation

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0 \quad (1)$$

Integrating over the control volume

$$\iint_w^e \frac{\partial u}{\partial x} dx dy + \iint_s^n \frac{\partial v}{\partial y} dy dx = 0 \quad (2)$$

$$(u_e - u_w)\Delta y + (v_n - v_s)\Delta x = 0 \quad (3)$$

Interpolating velocities at centroids instead of faces, considering linear interpolation gives,

$$(u_E - u_W)\Delta y + (v_N - v_S)\Delta x = 0 \quad (4)$$

As observed from Eq.4, that even we are solving for P node but there is no role of P node in the governing Equation which leads to a checker boarding problem.

Now, consider an unreal velocity field over a 2D domain,

10	5	10	5	10
5	10	5	10	5
10	5	10	5	10
5	10	5	10	5
10	5	10	5	10

Figure 3: Unreal velocity field

The continuity equation is still valid for this unreal, physically not existing field which is a drawback.

- Solution to above issue : Using Staggered grid we can overcome to this issue. This is not only solution to overcome this issue but one of the approaches available. In this approach we will take velocity over the faces instead of centroid of control volume.

Now, let us consider x momentum equation, (assuming incompressible flow and constant viscosity)

$$\rho(u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y}) = \mu(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}) - \frac{\partial P}{\partial x} \quad (5)$$

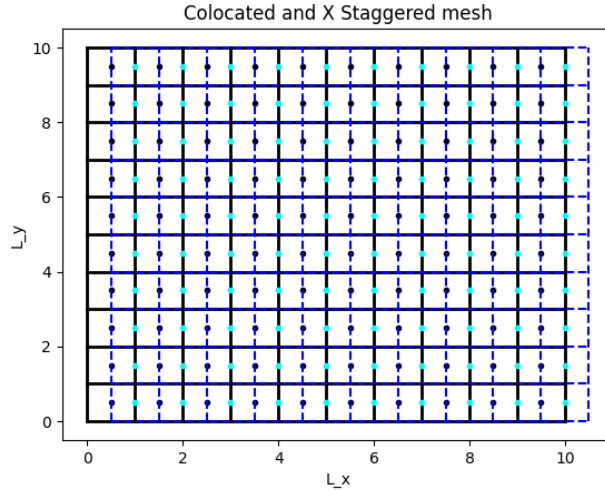


Figure 4: Staggered mesh in x direction over a collocated mesh

The blue dots represent the centroid at faces and black dots represent centroids at actual collocated grid.

Integrating the Eq.5 over the control volume from staggered mesh.

$$\iint_W^E u \frac{\partial u}{\partial x} dx dy + \iint_S^N v \frac{\partial u}{\partial y} dy dx = \frac{\mu}{\rho} \left(\iint_W^E \frac{\partial^2 u}{\partial x^2} dx dy + \iint_S^N \frac{\partial^2 u}{\partial y^2} dy dx \right) - \frac{1}{\rho} \iint_w^e \frac{\partial P}{\partial x} dx dy \quad (6)$$

After solving Eq.6 we get,

$$a_{eu} \cdot u_{i,j} = a_1 \cdot u_{i,j+1} + a_2 \cdot u_{i,j-1} + a_3 \cdot u_{i+1,j} + a_4 \cdot u_{i-1,j} + (P_{i,j} - P_{i,j+1}) \cdot \frac{\Delta y}{\rho} \quad (7)$$

where,

$$a_{ee} = \left(\frac{\mu \cdot \Delta y}{\rho \cdot \Delta x} - \frac{u_{i,j+1} \cdot \Delta y}{2} \right)$$

$$a_{ew} = \left(\frac{\mu \cdot \Delta y}{\rho \cdot \Delta x} + \frac{u_{i,j+1} \cdot \Delta y}{2} \right)$$

$$a_{en} = \left(\frac{\mu \cdot \Delta y}{\rho \cdot \Delta x} - \frac{v_{i+1,j} \cdot \Delta y}{2} \right)$$

$$a_{es} = \left(\frac{\mu \cdot \Delta y}{\rho \cdot \Delta x} + \frac{v_{i+1,j} \cdot \Delta y}{2} \right)$$

$$a_e = (a_{ee} + a_{ew} + a_{en} + a_{es})$$

Similarly, we can obtain expression for v using y momentum equation,

$$\rho \left(u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} \right) = \mu \left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} \right) - \frac{\partial P}{\partial y} \quad (8)$$

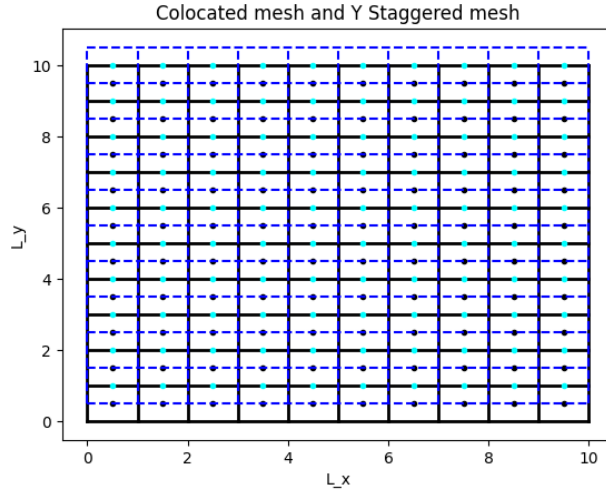


Figure 5: Staggered mesh in Y direction

Integrating the Eq.8 over the control volume from staggered mesh.

$$\iint_W^E u \frac{\partial v}{\partial x} dx dy + \iint_S^N v \frac{\partial v}{\partial y} dy dx = \frac{\mu}{\rho} \left(\iint_W^E \frac{\partial^2 v}{\partial x^2} dx dy + \iint_S^N \frac{\partial^2 v}{\partial y^2} dy dx \right) - \frac{1}{\rho} \iint_n^s \frac{\partial P}{\partial y} dy dx \quad (9)$$

$$a_{ev} \cdot v_{i,j} = a_5 \cdot v_{i,j+1} + a_6 \cdot v_{i,j-1} + a_7 \cdot v_{i+1,j} + a_8 \cdot v_{i-1,j} + (P_{i,j} - P_{i+1,j}) \cdot \frac{\Delta x}{\rho} \quad (10)$$

where,

$$\begin{aligned} a_{ee} &= \left(\frac{\mu \cdot \Delta x}{\rho \cdot \Delta y} - \frac{u_{i,j+1} \cdot \Delta x}{2} \right) \\ a_{ew} &= \left(\frac{\mu \cdot \Delta x}{\rho \cdot \Delta y} + \frac{u_{i,j+1} \cdot \Delta x}{2} \right) \\ a_{en} &= \left(\frac{\mu \cdot \Delta x}{\rho \cdot \Delta y} - \frac{v_{i+1,j} \cdot \Delta x}{2} \right) \\ a_{es} &= \left(\frac{\mu \cdot \Delta x}{\rho \cdot \Delta y} + \frac{v_{i+1,j} \cdot \Delta x}{2} \right) \\ a_e &= (a_{ee} + a_{ew} + a_{en} + a_{es}) \end{aligned}$$

3 SIMPLE algorithm

The steps involved in SIMPLE algorithm are

1. Guess a Pressure field (P^*) over the 2D domain.
2. Evaluate the velocity (u^*, v^*) using the guess pressure (P^*)
3. Evaluate pressure correction (P') using u^* and v^* .
4. Evaluate the corrected velocities (u', v') using P' .
5. Correct the velocities as well as pressure ($P = P^* + P'$, $u = u^* + u'$, $v = v^* + v'$)
6. Iterate the above four steps until get the convergence.

4 Results Obtained

We are using Gauss-Seidel method as our iterative method.

5 Validation with Benchmark results

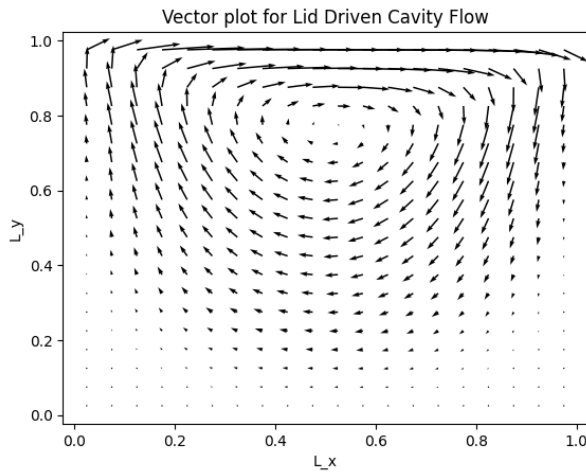


Figure 6: Vector plot obtained from OpenFoam

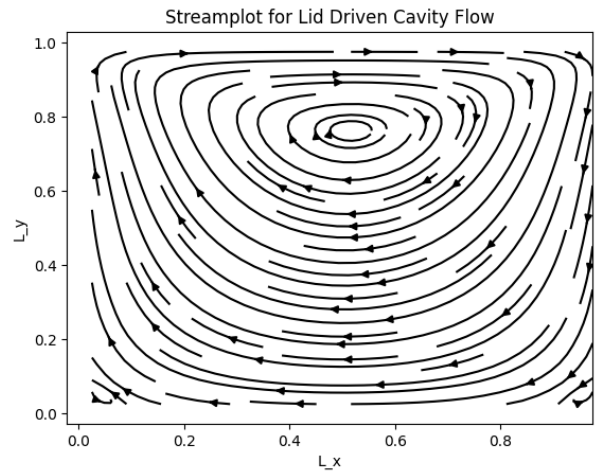


Figure 7: Streamline plot obtained from OpenFoam

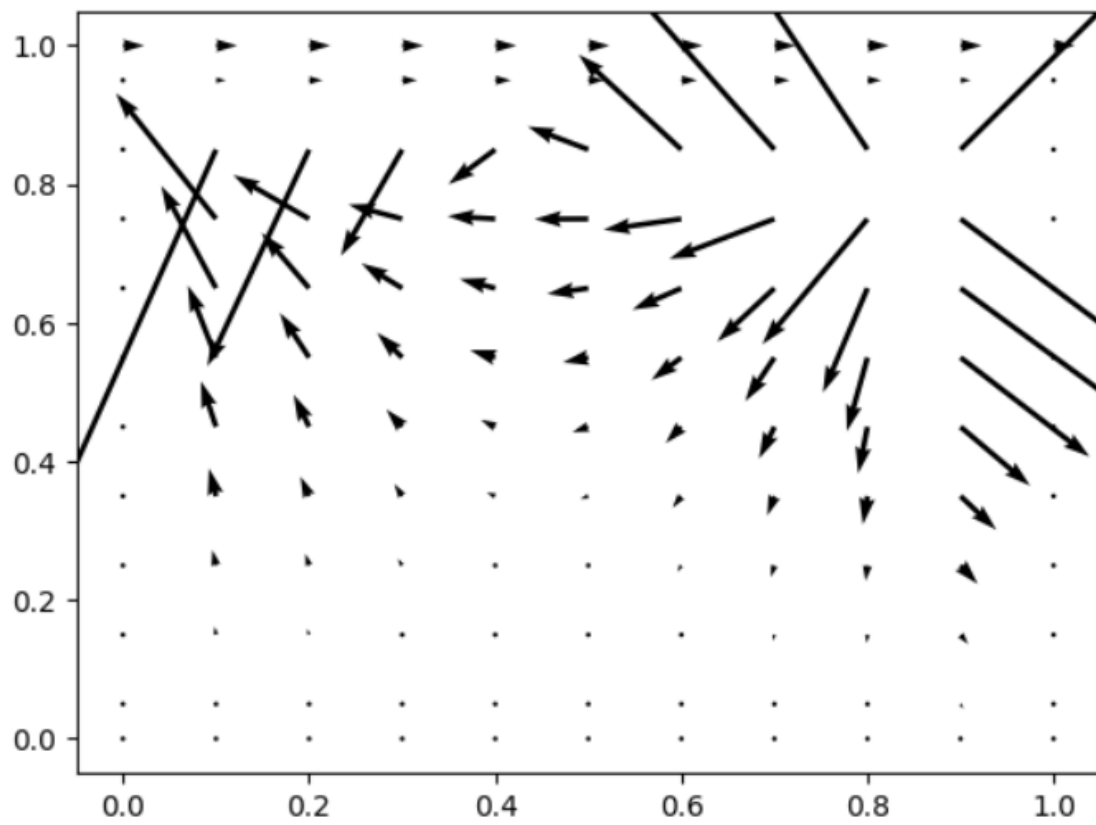


Figure 8: Results from our SIMPLE code